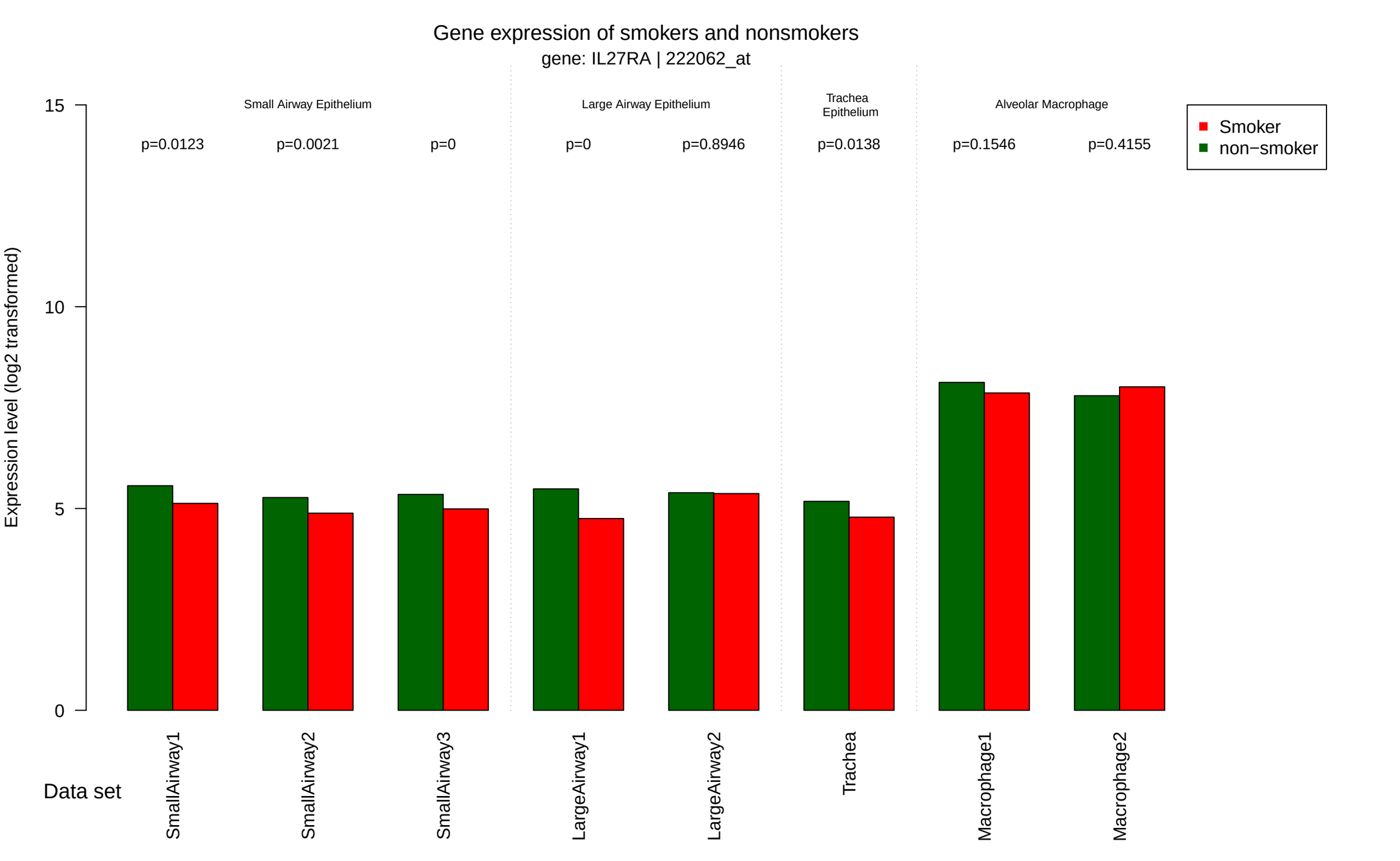
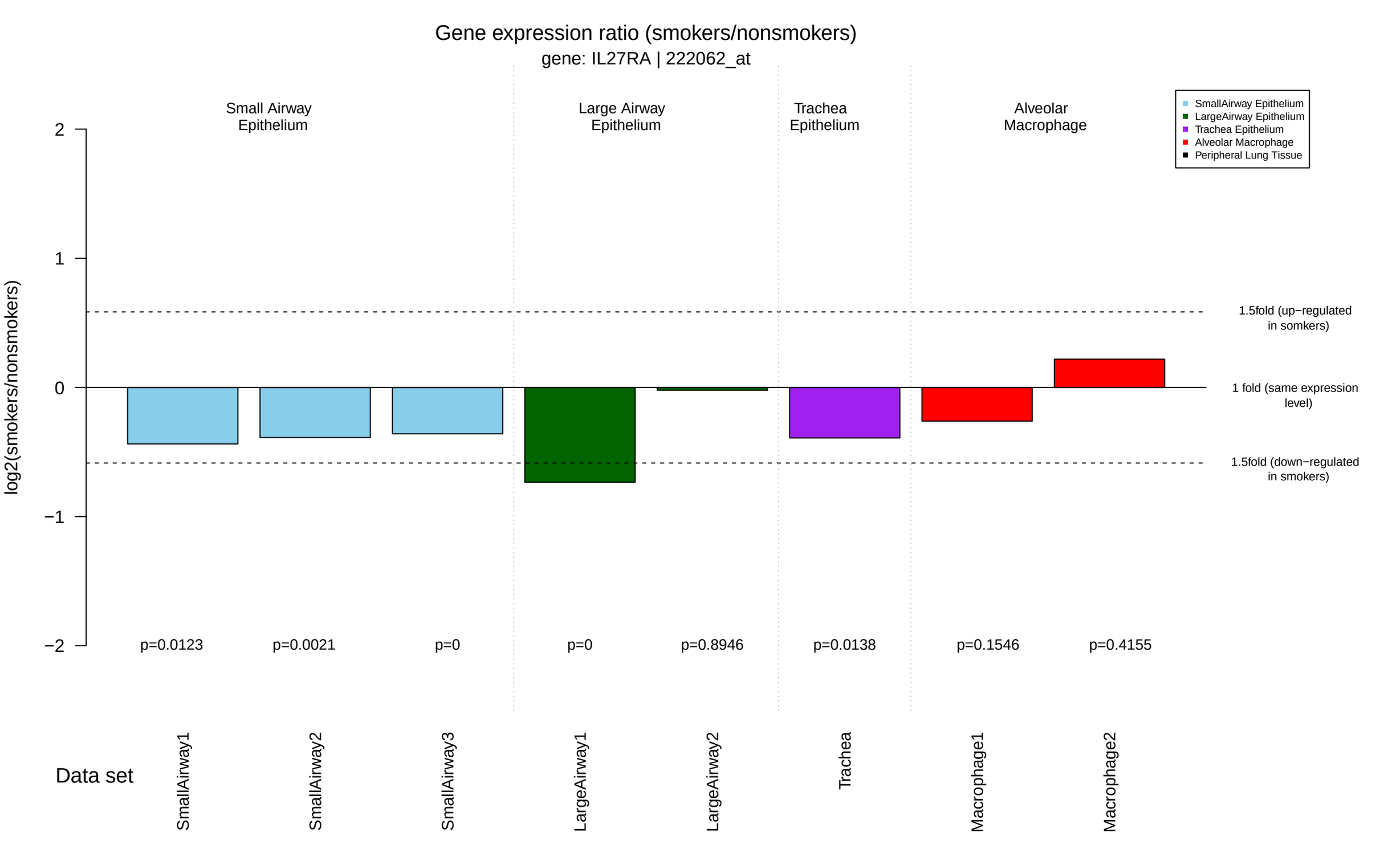
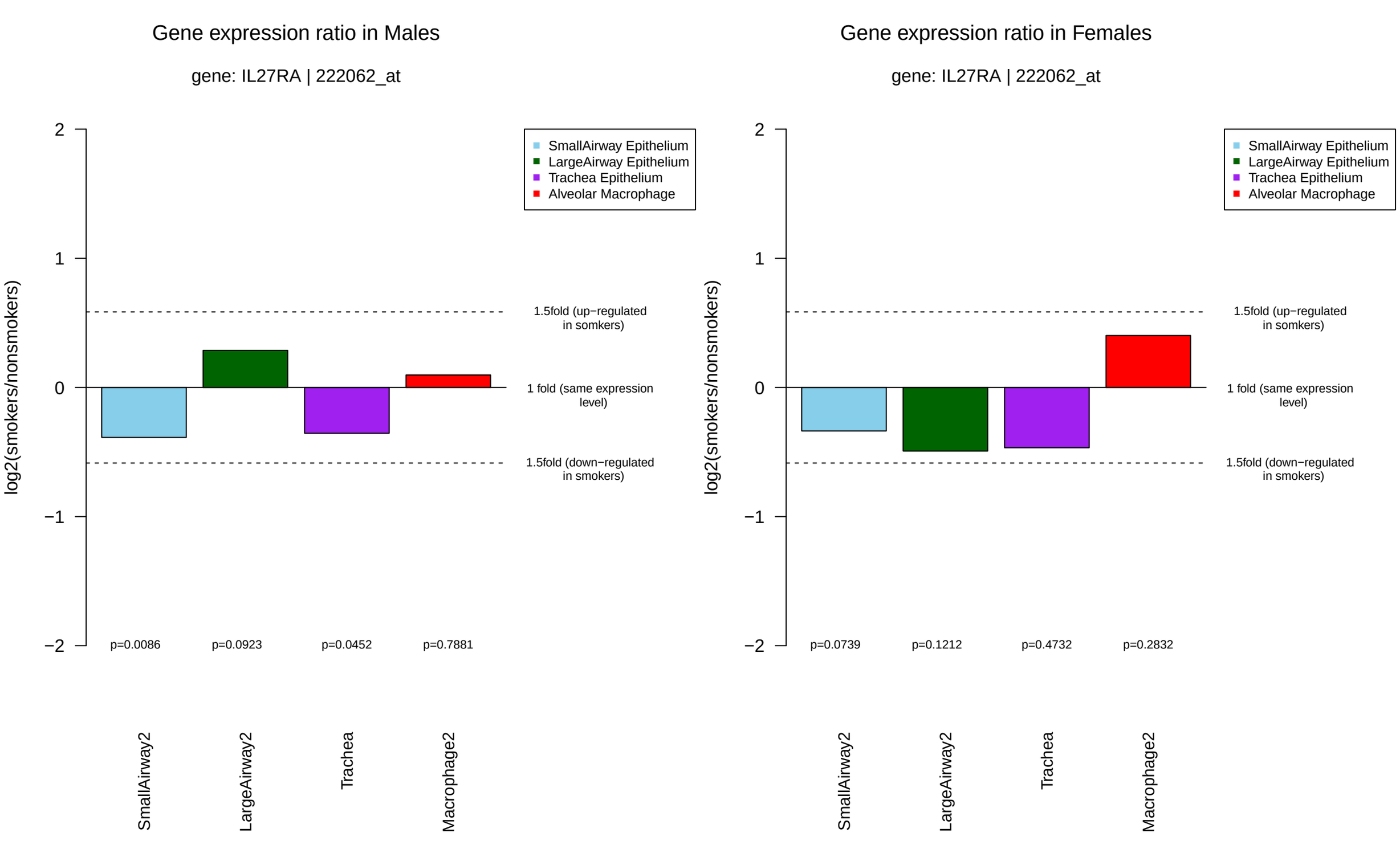
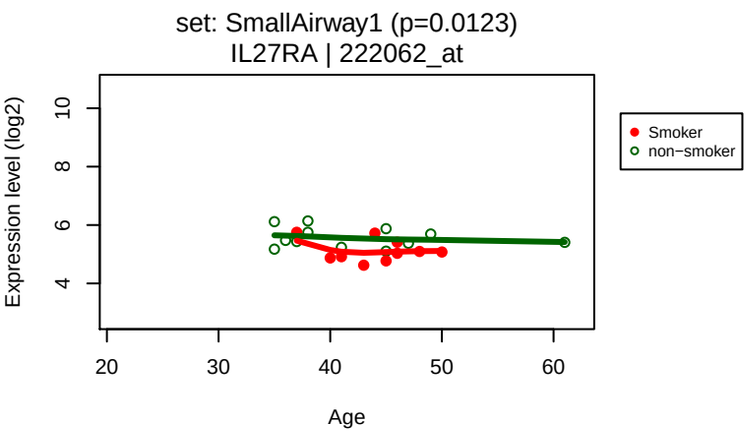
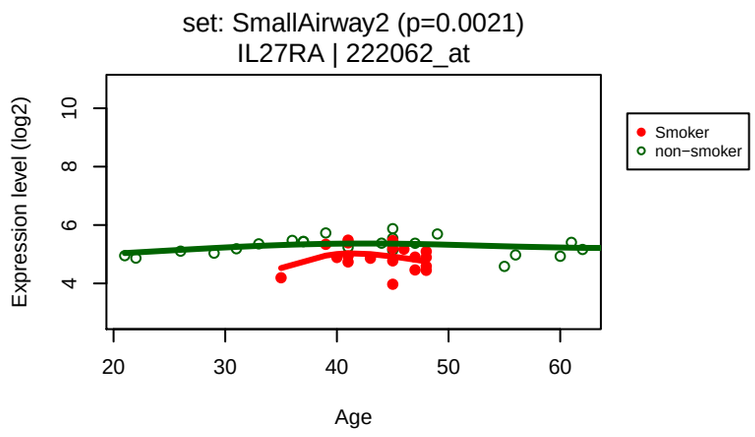
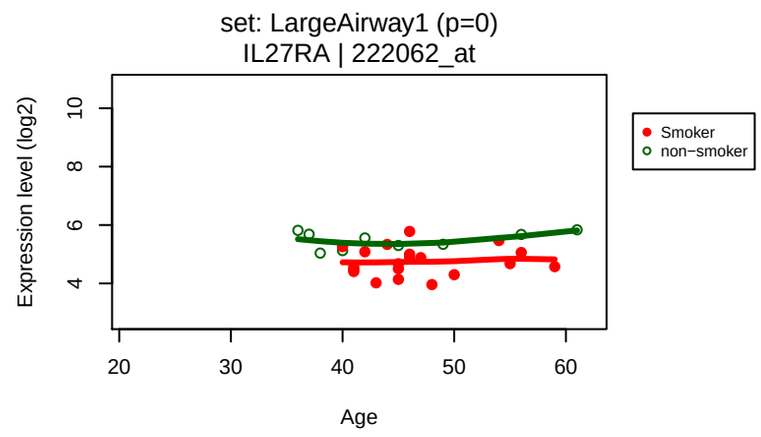

Gene expression of smokers and nonsmokers
gene: IL27RA | 222062_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0123
p=0.0021
p=0
p=0
p=0.8946
p=0.0138
p=0.1546
p=0.4155
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

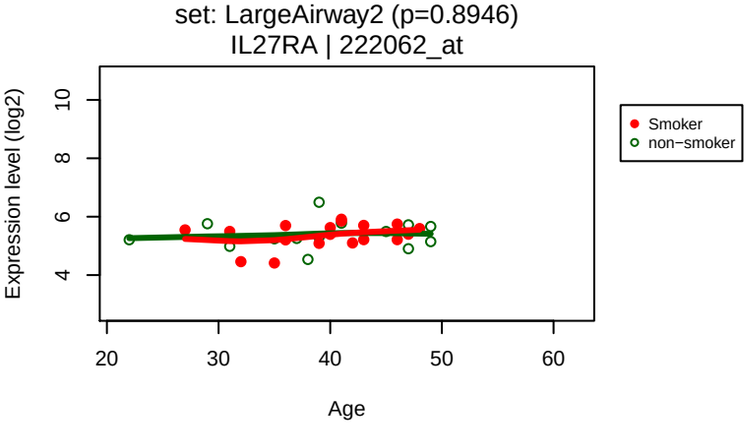

Gene expression ratio (smokers/nonsmokers)
gene: IL27RA | 222062_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0123
p=0.0021
p=0
p=0
p=0.8946
p=0.0138
p=0.1546
p=0.4155
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

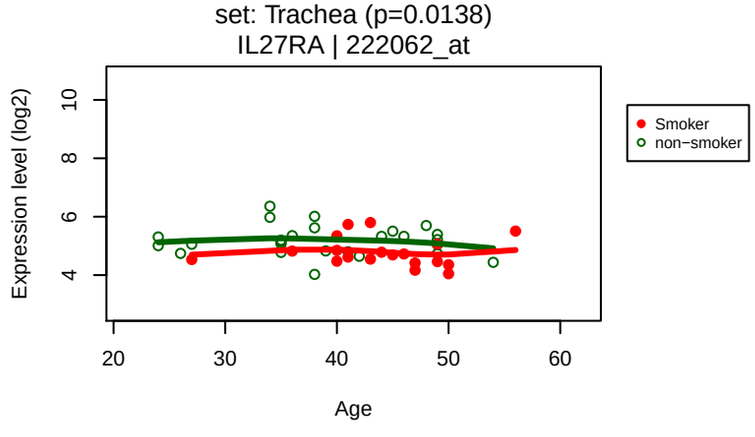

Gene expression ratio in Males
gene: IL27RA | 222062_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0086
p=0.0923
p=0.0452
p=0.7881
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL27RA | 222062_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0739
p=0.1212
p=0.4732
p=0.2832
SmallAirway2
LargeAirway2
Trachea
Macrophage2

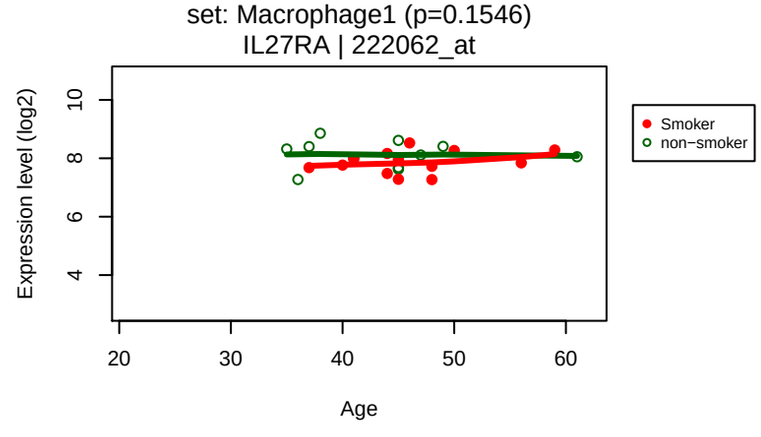

set: SmallAirway1 (p=0.0123)
IL27RA | 222062_at
Expression level (log2)
Age
Smoker
non−smoker

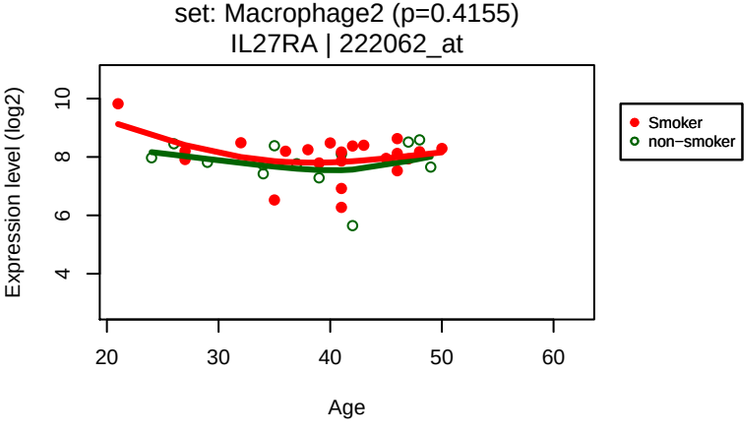

set: SmallAirway2 (p=0.0021)
IL27RA | 222062_at
Expression level (log2)
Age
Smoker
non−smoker

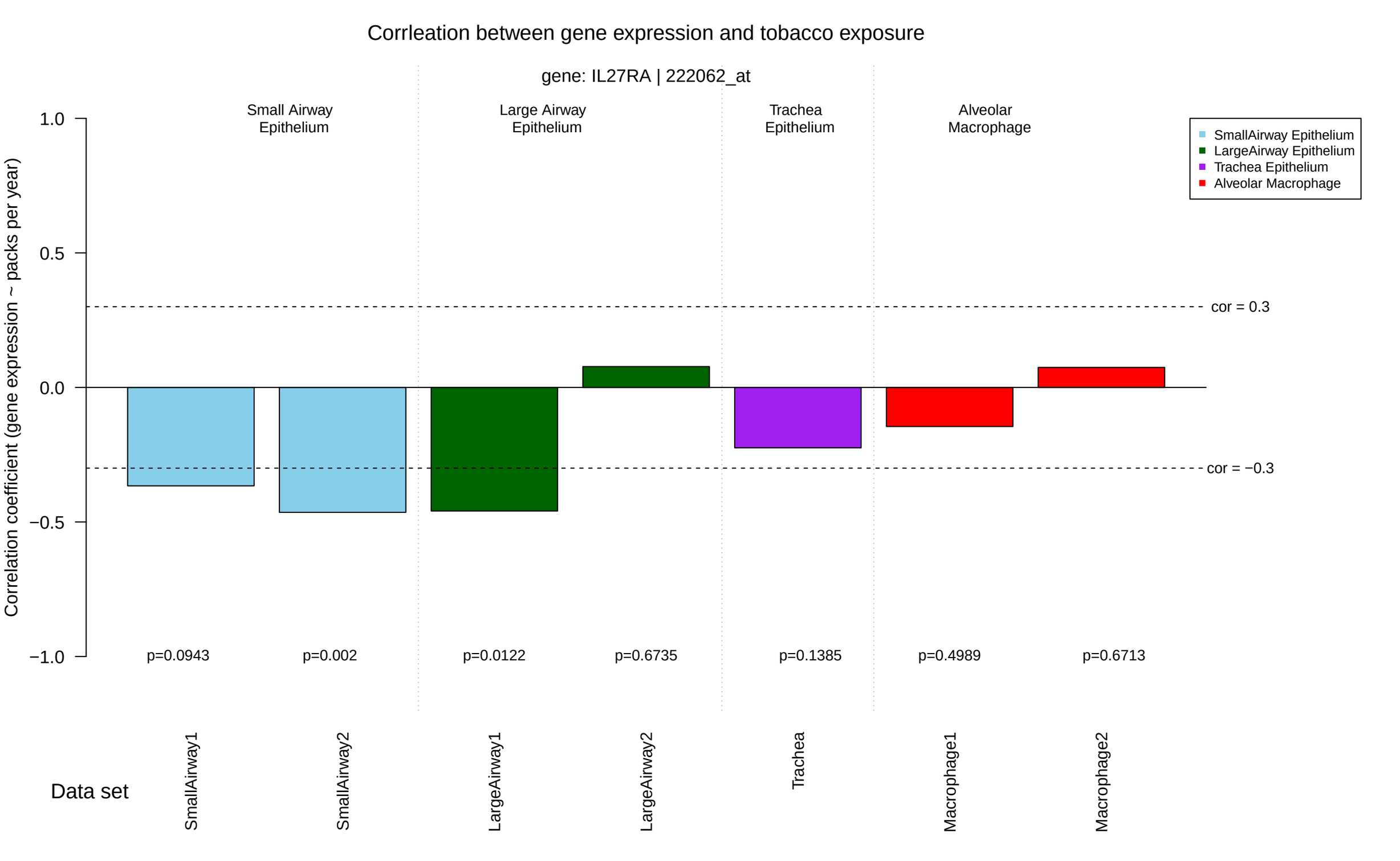

set: LargeAirway1 (p=0)
IL27RA | 222062_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8946)
IL27RA | 222062_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0138)
IL27RA | 222062_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1546)
IL27RA | 222062_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4155)
IL27RA | 222062_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL27RA | 222062_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0943
p=0.002
p=0.0122
p=0.6735
p=0.1385
p=0.4989
p=0.6713
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2